WORKSHEET

Chapter 6: Time series data

Use CAS or a spreadsheet to create a scatter plot of each table of time series data, then identify each graph as one or more of the following:

- negative secular trend
- positive secular trend
- seasonal pattern
- cyclic pattern
- random variation

1 To assist with planning, a census is taken four times a year of the population of a ski resort town.

Year	Population
Feb 2009	477
May 2009	580
Aug 2009	2728
Nov 2009	612
Feb 2010	478
May 2010	509
Aug 2010	3006
Nov 2010	715
Feb 2011	490
May 2011	578
Aug 2011	2944
Nov 2011	653
Feb 2012	502
May 2012	560
Aug 2012	3188
Nov 2012	699
Feb 2013	521
May 2013	568
Aug 2013	3192
Nov 2013	768

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2 Unemployment in the town of Bargon, 2012.

Month	No. of unemployed
January	678
February	645
March	672
April	641
May	628
June	638
July	615
August	618
September	593
October	581
November	599
December	573

3 Number of bands performing at a music festival in Sydney from 1992 until 2011.

Year	No. of bands performing
1992	21
1993	17
1994	25
1995	31
1996	44
1997	50
1998	41
1999	37
2000	37
2001	38
2002	35
2003	40
2004	49
2005	48
2006	48
2007	55
2008	41
2009	44
2010	66
2011	48

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4 Number of babies born with a low birthweight (under 2500 grams) in a suburban hospital.

Year	No. of low-birthweight babies born
1997	65
1998	81
1999	63
2000	56
2001	62
2002	57
2003	45
2004	70
2005	48
2006	55
2007	38
2008	67
2009	71
2010	59
2011	80
2012	49
2013	53

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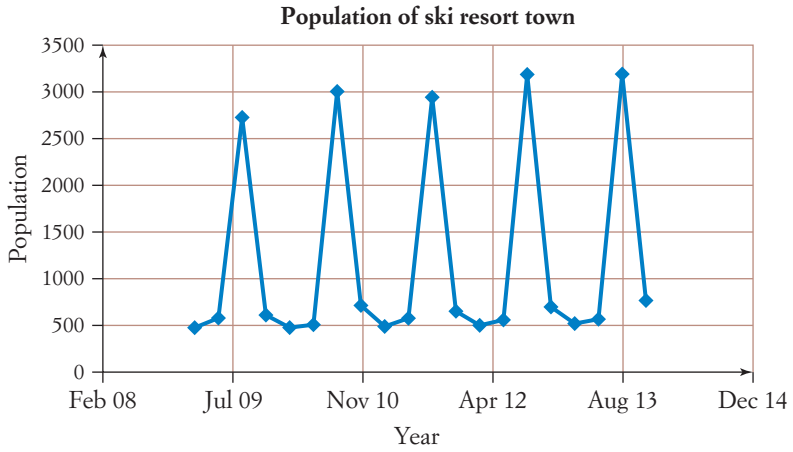
5 A real-estate agent keeps a record of the number of houses his business sells each year.

Year	Number of houses sold
1988	26
1989	35
1990	21
1991	29
1992	23
1993	16
1994	21
1995	19
1996	29
1997	15
1998	27
1999	36
2000	28
2001	35
2002	41
2003	21
2004	29
2005	21
2006	16
2007	32
2008	11
2009	19
2010	14
2011	29
2012	21
2013	33

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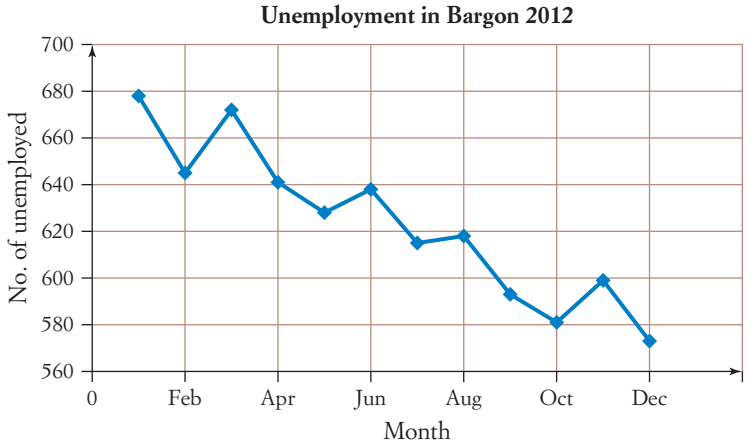
Answers

1



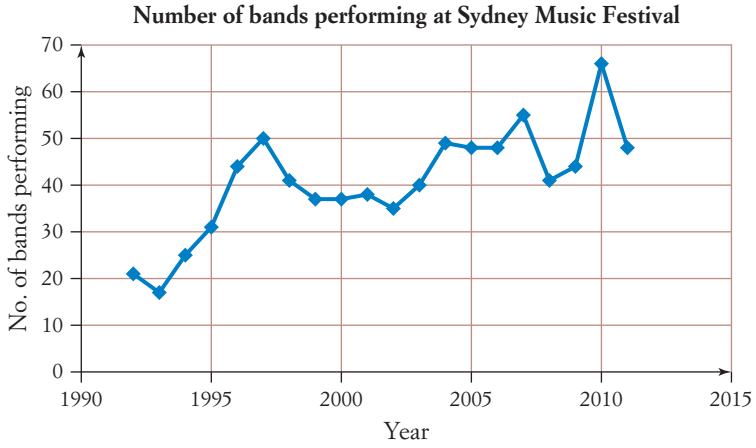
Cyclic, seasonal pattern, positive secular trend

2



Seasonal pattern, negative secular trend

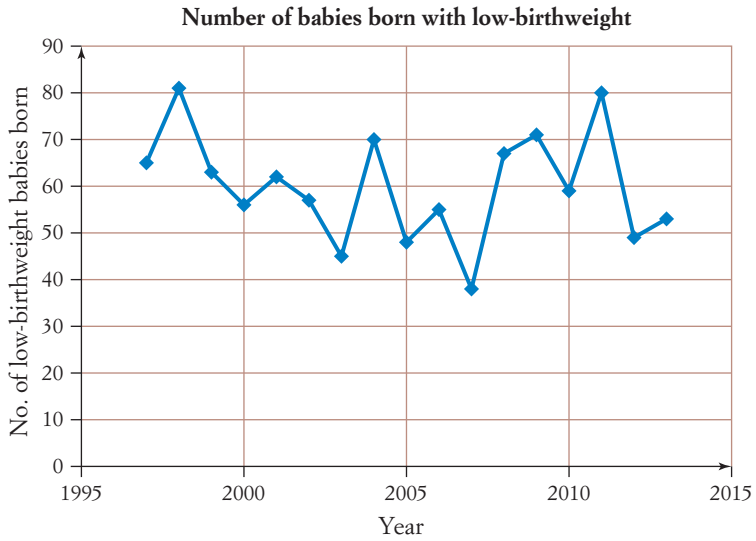
3



Seasonal pattern, positive secular trend

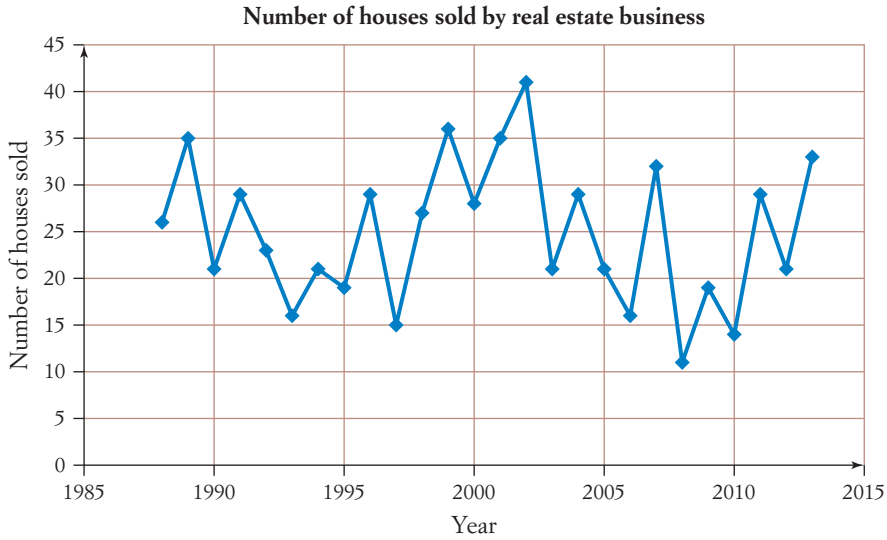
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4



Random variation

5



Cyclic pattern

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